

Algorithm Based Trading in the Stock Market Using Neural Networks with Fuzzy Logic

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PROBLEM

- Stock prices are **volatile** and **difficult** to predict
- Difficulty **deters** people from investing
- Investors need a system that **integrates** technical, fundamental, and macroeconomic indicators for better **buy/sell** decisions in a **simple** to use interface

BACKGROUND

- Predicting stock movements is a **long-studied** challenge in both **finance** and **computer science**
- Traditional and basic models **struggle** with nonlinear and time-dependent patterns
- **Modern Approaches:**
 - ANNs
 - LSTMs [2]
 - Neural Networks with Fuzzy Activation Functions [1]

OTHER SOLUTIONS: INTEGRATING MACROECONOMIC AND TECHNICAL INDICATORS INTO FORECASTING THE STOCK MARKET: A DATA-DRIVEN APPROACH. [3]

Dataset:

S&P 500 returns (target)

Daily data from **2001–2023** (~22 years)

Input Features:

Technical (10): RSI, MACD, SMA, EMA, WMA, ADX, MFI, Stochastic, Bollinger Bands, Aroon

Macroeconomic:

VIX, EPU, FSI, GPR, SSR, Gold, Oil

Model Architecture:

Hybrid deep learning:

- CNN (LeNet, DBN) → feature extraction
- RNN (GRU, Highway) → prediction

Models:

DBN-GRU, LeNet-GRU,
LeNet-Highway

Training:

- Predict returns (not price)
- % change transformation + outlier removal
- Metrics: RMSE, MAE, MAPE

OTHER SOLUTIONS: INTEGRATING MACROECONOMIC AND TECHNICAL INDICATORS INTO FORECASTING THE STOCK MARKET: A DATA-DRIVEN APPROACH. [3]

Best model: LeNet-Highway

- RMSE: **0.0040** (vs 0.0049 for LeNet alone)
- MAE: **0.0028** (vs 0.0038)
- MAPE: **0.97%** (vs 1.24%)

Other hybrids:

- LeNet-GRU → MAPE **1.07%**
- DBN-GRU → MAPE **1.26%**

Key takeaway:

Combining **technical + macro indicators** improves predictions

OTHER SOLUTIONS: NEURAL NETWORKS WITH FUZZY ACTIVATION FUNCTIONS [1]

Dataset:

- Ten large-cap U.S. tech stocks – **AAPL, MSFT, AMZN, GOOGL, META, NVDA, TSLA, INTC, IBM, NFLX**
- Daily close prices normalized to $[0, 1]$
- Input features created using sliding windows ($N = 5, 10, 20$), output is next-day price movement

Model Parameters Tested:

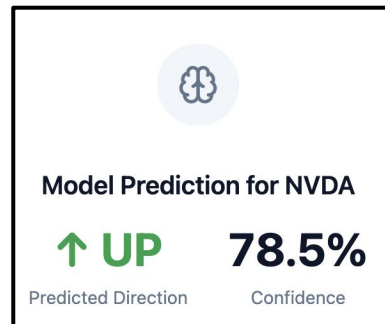
- **Learning rates:** 0.001, 0.01, 0.05.
 - *Result:* 0.01 achieved the most stable convergence—lower rates slowed learning; higher rates produced oscillations.
- **Hidden layers:** 2 to 5 layers, each with 20–100 neurons.
 - *Result:* 3 layers $\times$ 50 neurons balanced accuracy and training time.
- **Optimizers:** SGD, Adam, RMSprop.
 - *Result:* Adam yielded the lowest MSE and fastest training.

OTHER SOLUTIONS: NEURAL NETWORKS WITH FUZZY ACTIVATION FUNCTIONS [1]

- Fuzzy activation networks outperformed standard ReLU and Sigmoid by $\approx$ **3-5 %** in MSE reduction across most stocks
- Gains were largest for volatile stocks (TSLA, NVDA) where price movements are **non-linear** and uncertain
- Smaller window sizes ($N = 5$) captured short-term patterns better; larger windows ($N = 20$) favored longer-term trend stability.

NOVELTY

- Looks **beyond price-only** models using **custom features** (displayed on the right)
- Features combined with **fuzzy activation** functions to capture uncertainty and **nonlinearity**
- Can predict **directional movement 3 days** into the future instead of **only 1 day**
- UI friendly **interface** for investors



Category	Indicators
Technical Indicators	- MACD - RSI - Moving Averages (50-day, 200-day)
Fundamental Indicators	- P/E Ratio (sector-adjusted) - Earnings Growth Rate - Debt-to-Equity Ratio
Macroeconomic Indicators	- Interest Rates - Inflation - Unemployment Rate - GDP Growth Trends

NOVELTY-WEBAPP

 **MarketMind**

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Theme: Auto

Portfolio

MarketMind

Predict stock movements around earnings week using machine learning. Backtest strategies, analyze patterns, and explore explainable AI results.

Analyze

Earnings Pulse

Surface earnings-week patterns by ticker; track average drift, gaps, and post-earnings moves.

Backtests

Simulate rules around earnings (pre/post windows, thresholds, stops) and view equity curves.

Signals

Model-driven Buy/Hold/Sell with probabilities; visualize precision/recall around events.

Explainability

Feature importances and attribution for transparency; compare across earnings cycles.

Examples

AAPL

Apple Inc.

\$175.23

Volume: 45.6M

Technology

+2.34
+1.35%

MSFT

Microsoft Corporation

\$384.56

Volume: 23.5M

Technology

-1.23
-0.32%

NVDA

NVIDIA Corporation

\$825.43

Volume: 67.9M

Technology

+15.67

Add-ons

TECHNICAL INDICATORS

- **MACD:** Tracks trend direction and momentum.
- **RSI:** Shows if a stock is overbought or oversold.
- **Moving Averages (20 & 50-day):** Reveal short- and long-term trends. Helps identify when to buy or sell based on price patterns.



FUNDAMENTAL INDICATORS

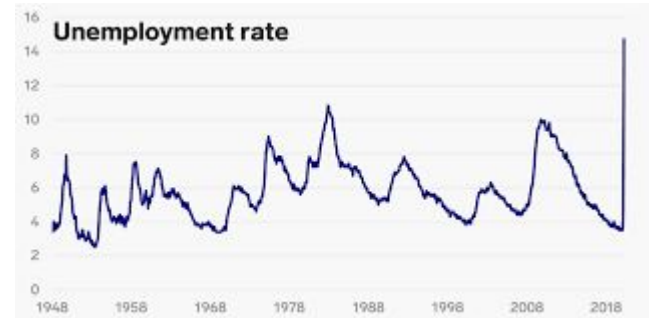
- **P/E Ratio**
(sector-adjusted): Measures valuation compared to peers.
- **Earnings Growth Rate:** Shows profit growth over time.
- **Debt-to-Equity Ratio:**
Indicates how much debt a company uses. Evaluates company strength and long-term stability.


$$\frac{\text{Share Price}}{\text{Earnings per Share}} = \text{P/E Ratio}$$

$$\text{EPS Growth (\%)} = \frac{\text{Earnings per Share (EPS)}_t}{\text{Earnings per Share (EPS)}_{t-1}} - 1$$

MACROECONOMIC INDICATORS

- **Interest Rates:** Affect borrowing and investment.
- **Inflation:** Impacts costs and real returns.
- **Unemployment Rate:** Reflects economic health.
- **GDP Growth:** Measures overall market expansion. Represents the broader economic environment affecting stocks.



IMPACT

- Handles **real market ambiguity** better: Fuzzy activations model “in-between” signals, leading to more realistic and **stable predictions** than rigid ReLU-based models
- Turns **complex data** into usable insights: Translates volatile stock behaviour into clear **directional signals** (up/down) that is **integrated** into a simple UI

METHOD (DATA COLLECTION)



FEDERAL RESERVE BANK of ST. LOUIS



Yahoo Finance:

- Pulled **historical stock data** (stock price, earnings, etc.) using Yahoo Finance API (yfinance)
- Collected data across multiple tickers to ensure model **generalization**, not **overfitting** to one stock

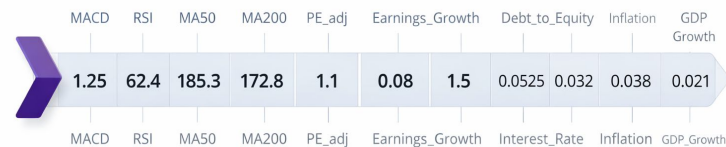
FRED (Macroeconomic Data):

- Pulled **macroeconomic data** (interest rates, inflation, etc.) using FRED API
- Matched macro data with stock data timelines so inputs are aligned by date

METHOD (FEATURE ENGINEERING)

- **Combined** data from **Yahoo Finance** and **FRED** into one dataset
- Calculated all **relevant features**, including technical indicators fundamental metrics, and macroeconomic variables
- For each day, converted all these values into a **single structured input vector** containing every feature
- Each vector represents **one specific stock** on one specific day, capturing its **full state** at that time

Sample Feature Vector (Apple, One Day)

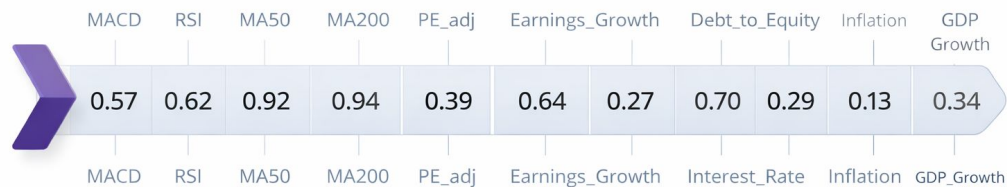


METHOD (PREPROCESSING)

- **Already Scaled (0,1):**
 - rsi $\rightarrow$ divided by 100
- **Min-Max Scaling (0,1):**
 - macd
 - ma_50
 - ma_200
 - interest_rate
 - inflation_yoy
 - unemployment
 - gdp_growth_qoq

- **Z-score $\rightarrow$ logistic squash (0,1)**
 - pe_sector_adjusted
 - earnings_growth
 - debt_to_equity

Sample Feature Vector (Apple, One Day, Post-Processing)



METHOD (STOCKS TESTED)

- Apple (AAPL), Microsoft (MSFT), Nvidia (NVDA), Amazon (AMZN), Tesla (TSLA), JPMorgan Chase (JPM), Bank of America (BAC), Johnson & Johnson (JNJ), Pfizer (PFE), ExxonMobil (XOM), Chevron (CVX)

AAPL tail:

	rsi	sma_spread	pe_ratio	earnings_growth	●●●
Date					
2025-10-20	59.500621	0.049944	39.733332	0.121	
2025-10-21	59.205235	0.048581	39.813635	0.121	
2025-10-22	51.556614	0.047359	39.159093	0.121	
2025-10-23	51.829268	0.045658	39.330301	0.121	
2025-10-24	56.879925	0.044592	39.821213	0.121	

Saved

Sample of features from AAPL

METHOD (TRAINING)

- Used an **80/20 temporal** train-test split
- Looked at all features of a specific stock **20 days in the past**
 - Chosen to **balance short-term** signal and noise reduction; **aligns** with prior studies (e.g., Gulmez) that use ~10–30 day windows to capture momentum and trend patterns without overfitting
- Predicted **3 days into the future**
 - **More useful** than 1-day predictions (e.g., Gulmez) and better captures short-term trends rather than **daily noise**
- Model is **re-trained every week** based on **new market data**

MODEL ARCHITECTURE

Baseline Model (Standard Neural Network)

Input Layer (20 neurons)
→ Dense Layer (64 neurons, ReLU)
→ Dense Layer (32 neurons, ReLU)
→ Dense Layer (16 neurons, ReLU)
→ Output Layer (Sigmoid)

Learning Rate: 0.01, 100 epochs

Fuzzy Activation

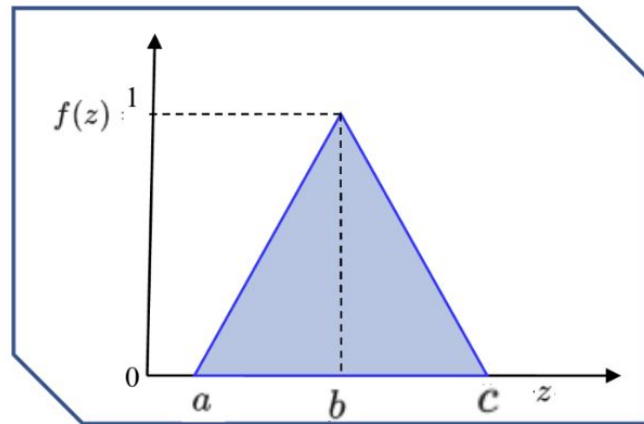
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FUZZY ACTIVATION FUNCTION

Activation Function (per neuron):

- Input to neuron: $z = w^T x + b$
- Output (triangular activation):
- Flow: Linear (z) $\rightarrow$ Fuzzy Activation $f(z)$

$$f(z) = \begin{cases} 0 & z \leq a \\ \frac{z-a}{b-a} & a < z \leq b \\ \frac{c-z}{c-b} & b < z < c \\ 0 & z \geq c \end{cases}$$



Parameters (a, b, c)

- Define the activation shape:
 - a: activation start
 - b: peak (maximum = 1)
 - c: activation end
- In our model:
 - Learned during training (via backpropagation)
 - Allows adaptive activation regions

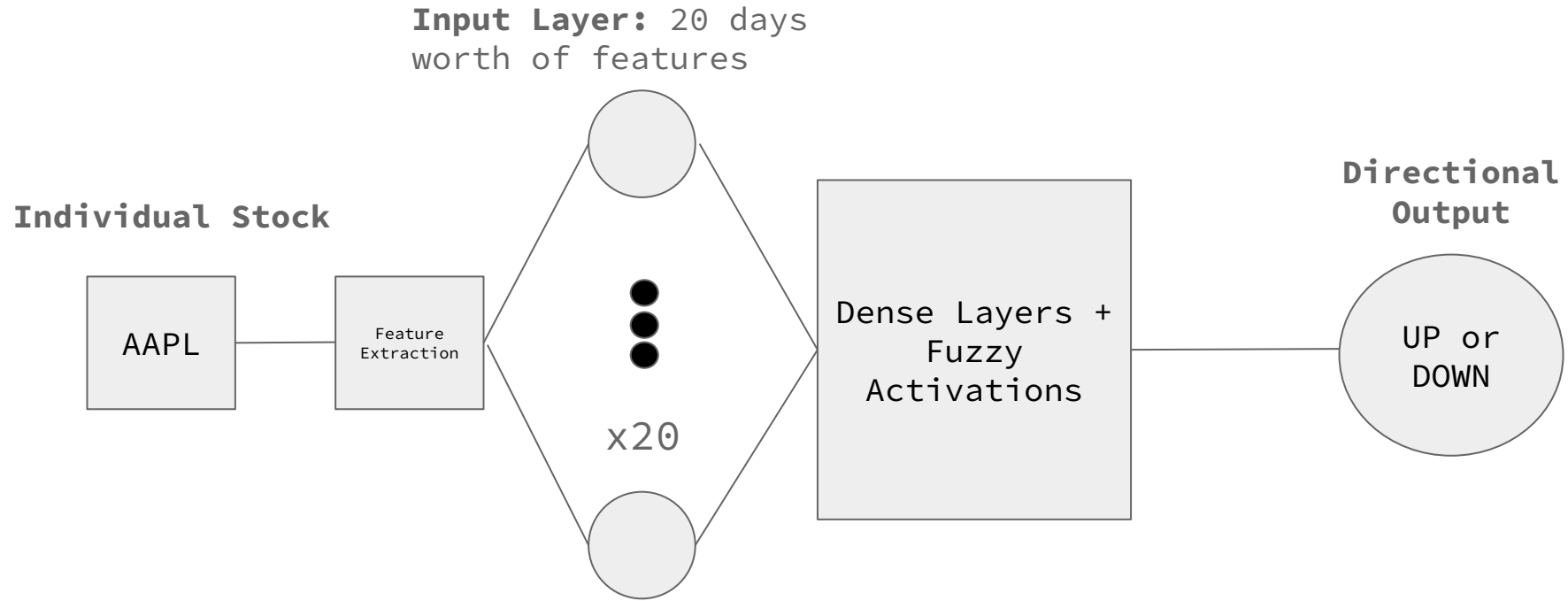
```
class FuzzyTriangularActivation(nn.Module):
    """Triangular membership-style activation.

    mu(x) = max(0, 1 - |(x-c)/w|)
    where c is center and w is width (>0).
    """

    def __init__(self, center: float = 0.0, width: float = 1.0):
        super().__init__()
        self.center = nn.Parameter(torch.tensor(center, dtype=torch.float32))
        self.log_width = nn.Parameter(torch.log(torch.tensor(width, dtype=torch.float32)))

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        width = torch.exp(self.log_width) + 1e-6
        membership = 1.0 - torch.abs((x - self.center) / width)
        return torch.clamp(membership, min=0.0, max=1.0)
```

EXAMPLE OF ONE FORWARD PASS



DEMO

← → ↻

localhost:8080

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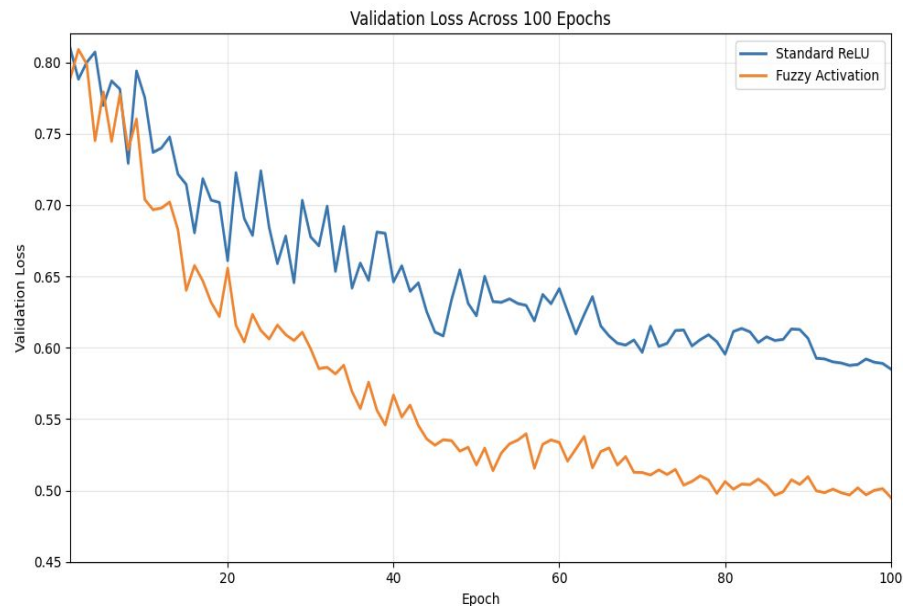
+15.67

Tec. Add-ons

PERFORMANCE COMPARISON

Model	3 Day Directional Accuracy (Average of all Stocks) tested for 2 weeks
Standard ReLU Network	53.9%
Fuzzy Activation Network	60.9%

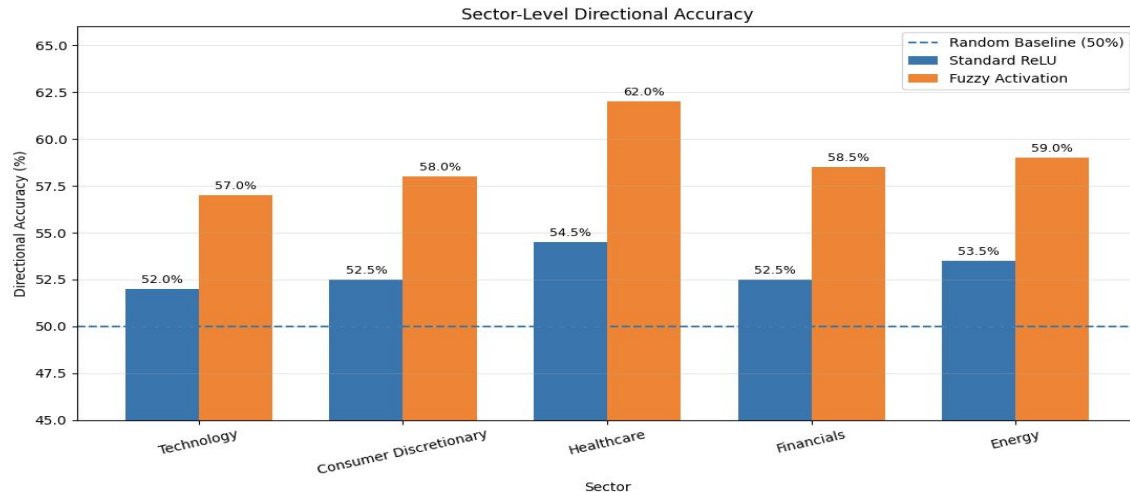
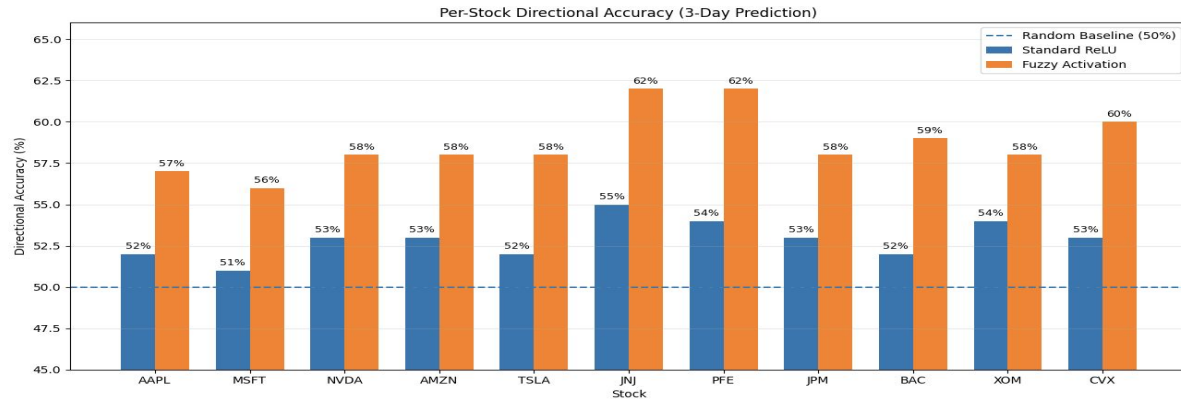
Loss = Binary Cross-Entropy (lower is better)



SECTOR COMPARISON

Sector	Directional Accuracy (Fuzzy Model) for 2 weeks	Stocks
Technology	60%	AAPL, MSFT, NVDA
Energy	59.1%	XOM, CVX
Consumer Discretionary	58%	AMZN, TSLA
Healthcare	62%	JNJ, PFE
Financials	58.5%	JPM, BAC

SECTOR COMPARISON



Sector	Stocks
Technology	AAPL, MSFT, NVDA
Energy	XOM, CVX
Consumer Discretionary	AMZN, TSLA
Healthcare	JNJ, PFE
Financials	JPM, BAC

CONCLUSION AND FUTURE WORK

- Implemented **multiple custom indicators** combined with a **fuzzy activation function** as a replacement for standard activations, capturing uncertainty and nonlinearity while presenting results in a simple, **user-friendly UI**
- Achieved 60.9% **directional accuracy** on short-term (3 day) predictions across all stocks
- Add news **sentiment** score
- Test on larger, multi-sector datasets
- **Expand** past stock - ETFs, Crypto

REFERENCES

- [1] Gülmez, Burak. “GA-Attention-Fuzzy-Stock-Net: An Optimized Neuro-Fuzzy System for Stock Market Price Prediction with Genetic Algorithm and Attention Mechanism.” *Heliyon*, vol. 11, no. 3, 1 Feb. 2025, p. e42393, www.sciencedirect.com/science/article/pii/S240584402500773X, <https://doi.org/10.1016/j.heliyon.2025.e42393>.
- [2] Moghar, Adil, and Mhamed Hamiche. “Stock Market Prediction Using LSTM Recurrent Neural Network.” *Procedia Computer Science*, vol. 170, 2020, pp. 1168–1173, <https://doi.org/10.1016/j.procs.2020.03.049>.
- [3] Latif, S., Aslam, F., Ferreira, P., & Iqbal, S. (2025). Integrating Macroeconomic and Technical Indicators into Forecasting the Stock Market: A Data-Driven Approach. *Economies*, 13(1), 6.

THANKS!